

(14)

EE621: Computer Vision
Quiz 1

Full Marks: 20

Duration: 40 mins

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1. Find the Euclidian, City block, and Chessboard distances, respectively, between the two points $A(3, -4)$ and $B(-5, 6)$.

Euclidian distance $d = 12.806$

City Block distance $= 18$

Chessboard distance $= 10$

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2. Consider a point $P(45, -30, 5)$ in projective space. Convert the point into 2D real coordinates and determine whether the point lies at infinity.

(a) $(8, 16)$ and point does not lie at infinity.

☒ (b) $(9, -6)$ and point does not lie at infinity.

(c) $(8, -16)$ and point at infinity.

(d) $(3, -4)$ and point at infinity.

2

3. Identify the transformation represented by the following matrix:

$$\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) Rotation

(b) Y-shear

☒ (c) X-shear

(d) Reflection

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4. A point $(6, 1)$ is rotated 60° clockwise, then scaled by a factor of 4, and finally translated by $(-2, 3)$. Find the final coordinates of the point.

~~$(-5.53, -15.78)$~~ $(13.464, -15.784)$

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5. A point $P(2, -1)$ undergoes the following transformations in order:

1. Rotation of 60° anticlockwise about the point $(1, 1)$
2. Reflection about the line $y = -x$
3. Non-uniform scaling with $S_x = 2$, $S_y = 3$ about the origin
4. Translation by vector $(3, -2)$

Using homogeneous coordinates, the final coordinates of the point are:

~~$(-0.53, -15.78)$~~

- (a) $(5 + \sqrt{3}, -4 + 2\sqrt{3})$
 (b) $(3 + 2\sqrt{3}, -5 + \sqrt{3})$
 (c) $(3 + \sqrt{3}, -4 + 2\sqrt{3})$
 (d) $(5 + 2\sqrt{3}, -4 + \sqrt{3})$

6. For a Lambertian surface with albedo ρ , the BRDF is:

- (a) ρ
 (b) $\rho \cos \theta$
 (c) $\frac{\rho}{\pi}$
 (d) $\pi \rho$

7. In a calibrated stereo setup, the focal length is 800 pixels and the baseline is 0.2 m. If the disparity of a point is 40 pixels, what is its depth?

- (a) 2 m
 (b) 4 m
 (c) 8 m
 (d) 16 m

8. A surface has reflectance $\rho = 0.5$. Incident light makes an angle of 60° with the surface normal and has an intensity of 40 W/m^2 along the normal direction.

Assuming the surface follows Lambertian reflection, find the reflected radiance.

- (a) $3.18 \text{ W/m}^2/\text{sr}$
 (b) 4.25 W/m^2
 (c) 3.18 W/m^2
 (d) $4.25 \text{ W/m}^2/\text{sr}$

9. Find Radon transform for $\theta = 0^\circ$?

$$f(x, y) = \begin{cases} A, & x^2 + y^2 \leq r^2 \\ 0, & \text{otherwise} \end{cases}$$

(b)

$$\int_{-\sqrt{r^2 - p^2}}^{\sqrt{r^2 - p^2}} A \, dy$$

(c)

$$\int_{-\sqrt{r^2 - 2p^2}}^{\sqrt{r^2 - 2p^2}} A \, dy$$

(d)

$$\int_{-\sqrt{r^2}}^{\sqrt{r^2 - p^2}} A \, dy$$

$$\int_{-\sqrt{r^2}}^{\sqrt{r^2}} A \, dy$$

10. Let $f(x, y) \in L^2(\mathbb{R}^2)$. The 1D Fourier transform (with respect to s) of its Radon transform satisfies:

- (a) $F^*(\omega, \theta)$
 (b) $F(\omega \cos \theta, \omega \sin \theta)$
 (c) $F(\cos \theta, \sin \theta)$
 (d) $F(\omega \sin \theta, \omega \cos \theta)$

$$\mathcal{F}_s\{R_f(s, \theta)\} =$$